



Modeling and control of Voltage Source Converters (VSCs) Part II

SUNRISE Summer School

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This project has received funding from the European Union's Horizon Widening participation and spreading excellence programme under grant agreement No. 101079200





Outline

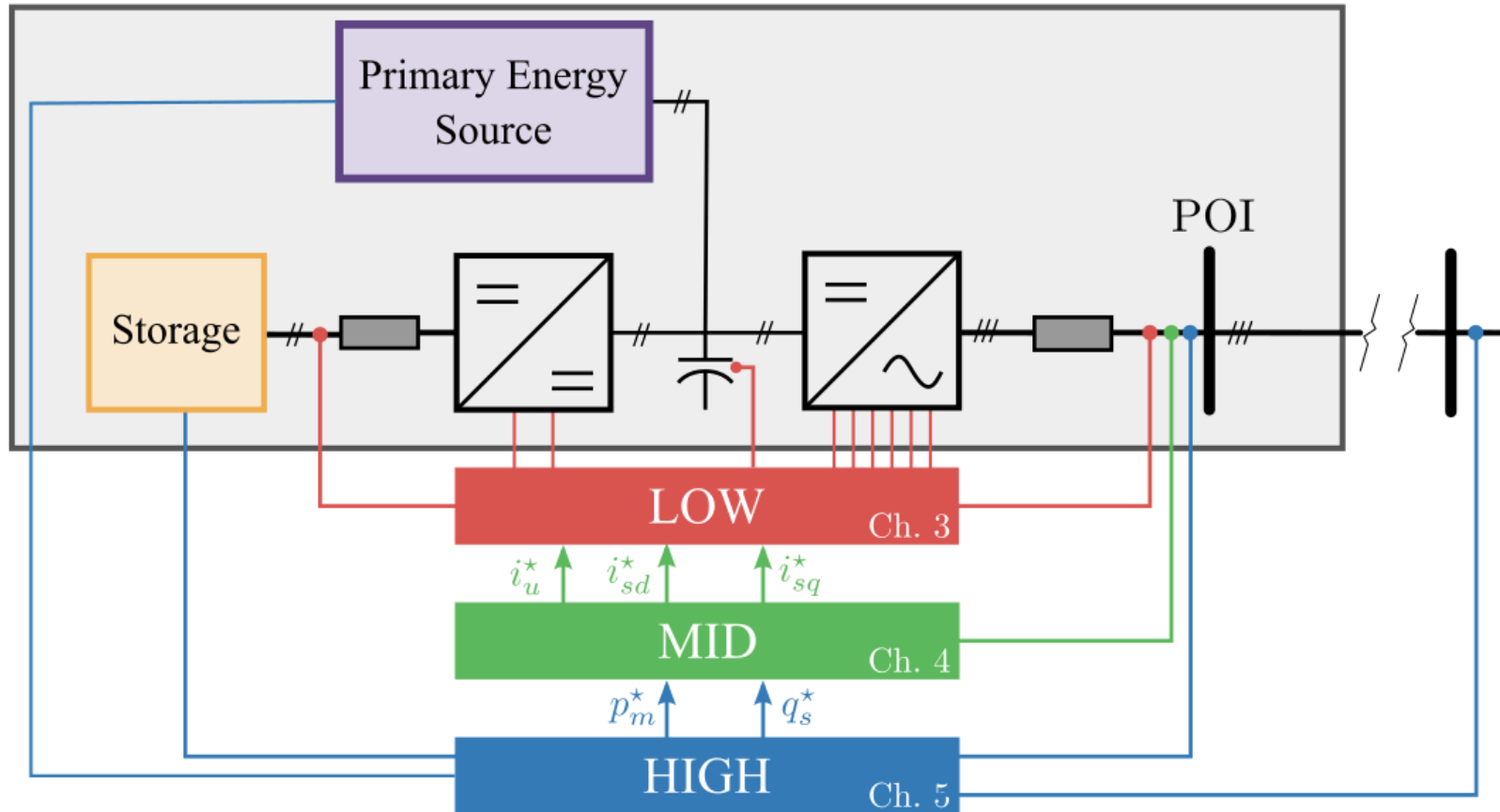
- Introduction
- Three-phase inverter
- Pulse Width Modulation (PWM)
- Coupling filter
- Averaged model of Three-Phase Three-wire Two-level VSC in abc
- Averaged model of Three-Phase Three-wire Two-level VSC in dq
- Grid synchronization: Phase Lock Loop (PLL)
- VSC control
- Current control in dq coordinates
- Computation of reference current
- DC-link voltage control



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VSC control

- Hierarchical controller of the VSCs



LOW LEVEL CONTROL

- PWM and **current controller**

MID LEVEL CONTROL

- **P-Q Control: GFL** or GFM
- **DC bus voltage control**

HIGH LEVEL CONTROL

- Primary frequency control
- Voltage control
- Power Smoothing
- EMS of ESS



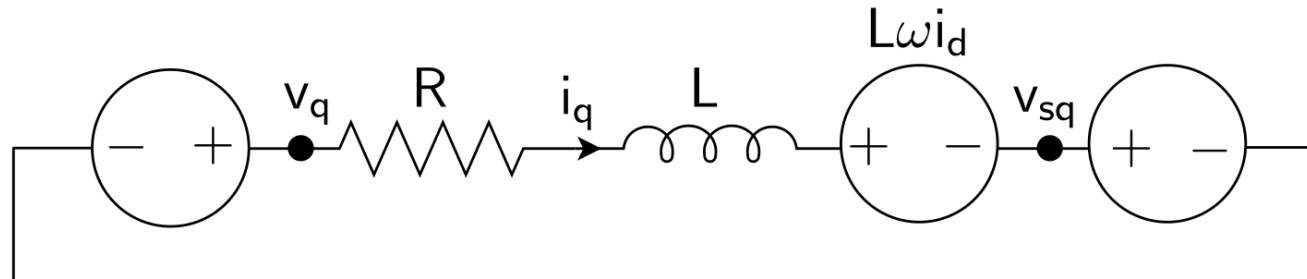
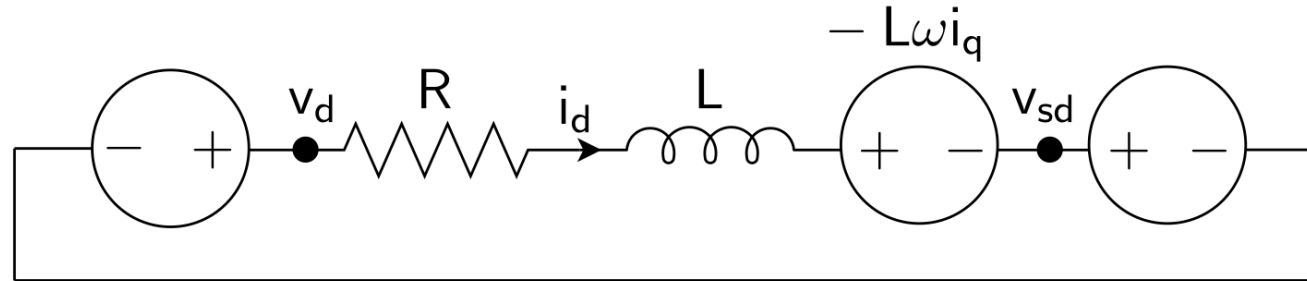


Current control in dq coordinates

- AC-side equivalent model in dq:

$$\frac{di_d}{dt} = \frac{1}{L} \left(\eta_d \frac{v_{dc}}{2} + L\omega i_q - Ri_d - v_{sd} \right)$$

$$\frac{di_q}{dt} = \frac{1}{L} \left(\eta_q \frac{v_{dc}}{2} - L\omega i_d - Ri_q - v_{sq} \right)$$





Current control in dq coordinates

Auxiliary variables: u_d and u_q

$$\frac{di_d}{dt} = \frac{1}{L} \left(\underbrace{\eta_d \frac{V_{dc}}{2} + L\omega i_q - v_{sq}}_{u_d} - Ri_d \right)$$
$$\frac{di_q}{dt} = \frac{1}{L} \left(\underbrace{\eta_q \frac{V_{dc}}{2} - L\omega i_d - v_{sd}}_{u_q} - Ri_q \right)$$



$$u_d = \eta_d \frac{V_{dc}}{2} + L\omega i_q - v_{sd}$$

$$u_q = \eta_q \frac{V_{dc}}{2} - L\omega i_d - v_{sq}$$



System to be controlled

$$\frac{di_d}{dt} = \frac{1}{L} (u_d - Ri_d)$$

$$\frac{di_q}{dt} = \frac{1}{L} (u_q - Ri_q)$$



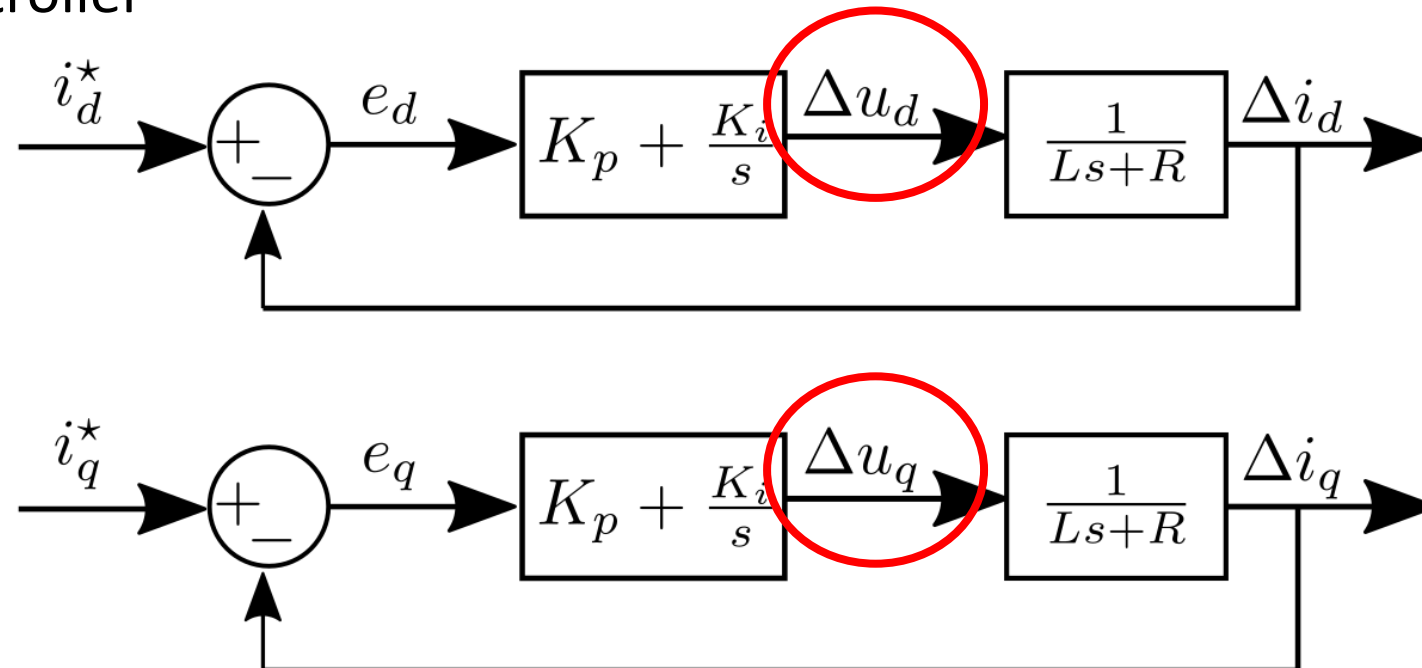


Current control in dq coordinates

Transfer function to be controlled



PI current controller



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Current control in dq coordinates

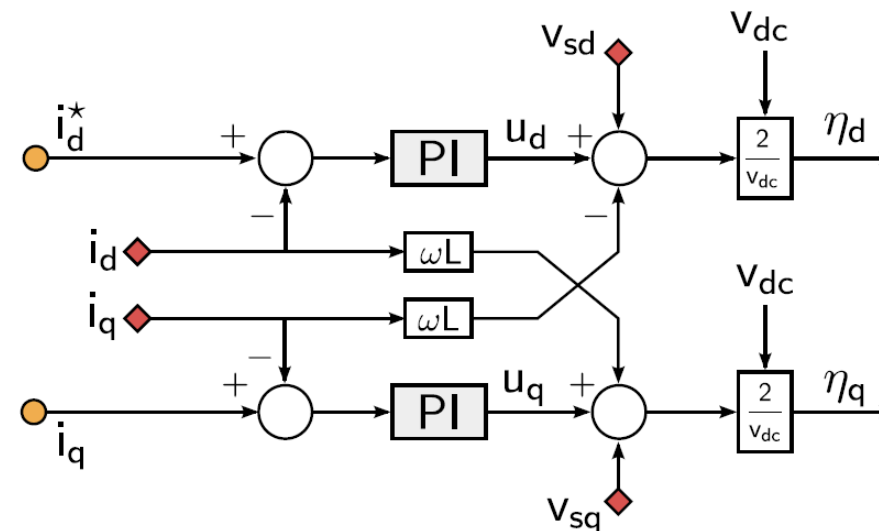
Relationship auxiliary and actual inputs Actual Inputs computation: η_d and η_q

$$u_d = \eta_d \frac{V_{dc}}{2} + L\omega i_q - v_{sd}$$

$$u_q = \eta_q \frac{V_{dc}}{2} - L\omega i_d - v_{sq}$$

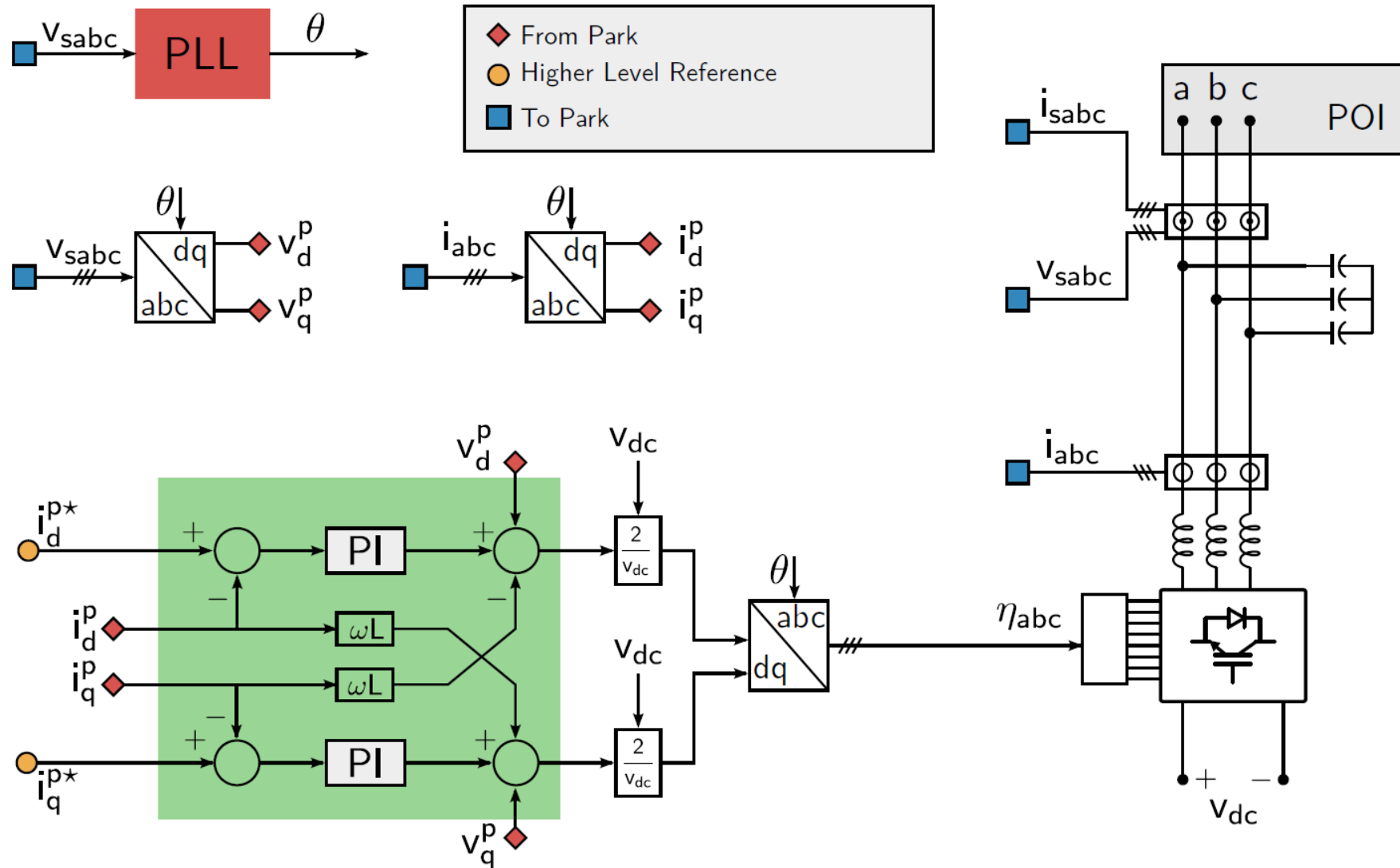
$$\eta_d = \frac{2}{V_{dc}} (u_d - L\omega i_q + v_{sd})$$

$$\eta_q = \frac{2}{V_{dc}} (u_q + L\omega i_d + v_{sq})$$





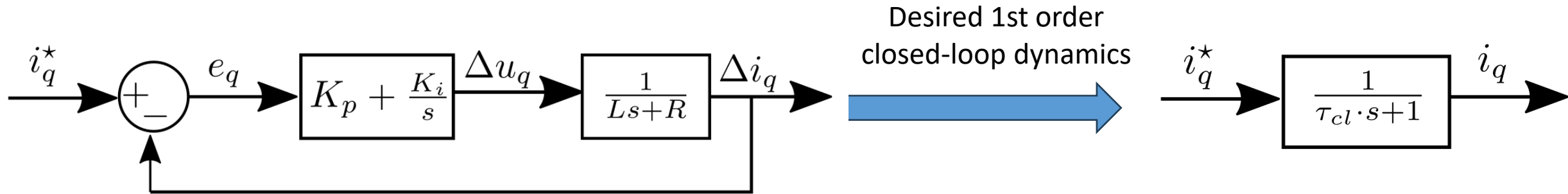
Current control in dq coordinates



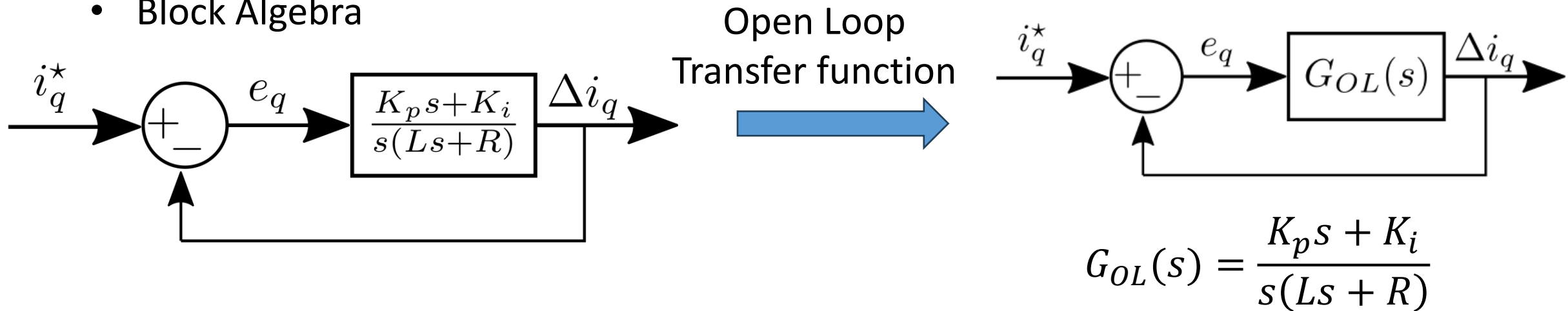


Current control in dq coordinates

Tuning of the controller gains



- Block Algebra

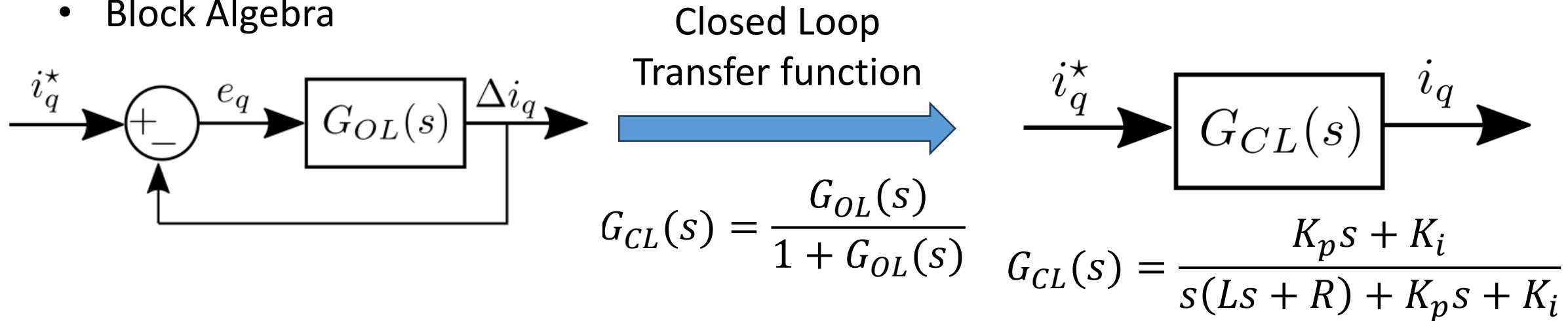




Current control in dq coordinates

Tuning of the controller gains

- Block Algebra



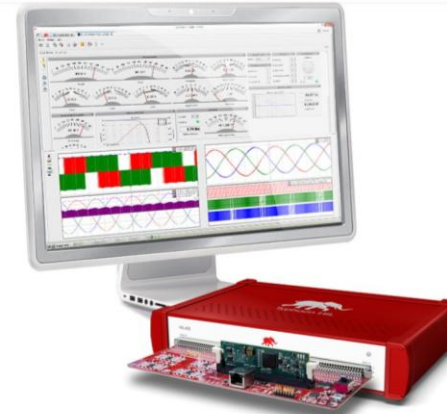
- Identifying $G_{CL}(s)$ with the desired closed-loop dynamic

$$G_{CL}(s) = \frac{1}{s \left(\frac{Ls + R}{K_p s + K_i} \right) + 1} = \frac{1}{\tau_{cl} s + 1} \longrightarrow Ls + R = \tau_{cl} (K_p s + K_i)$$

$$K_p = \frac{L}{\tau_{cl}} ; K_i = \frac{R}{\tau_{cl}}$$



Methodology



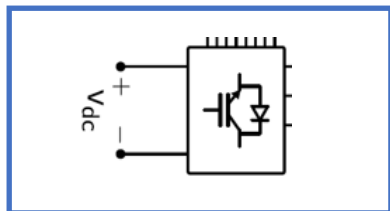
Study of the state of the art
and development

Modelling and validation in
off-line simulation

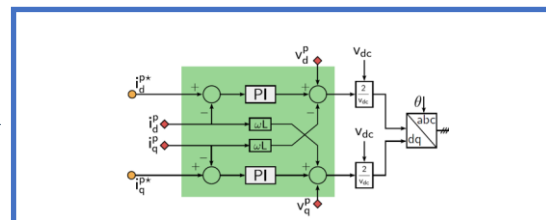
Modelling and validation in
Control Hardware in the Loop

Validation in
Experimental Platform

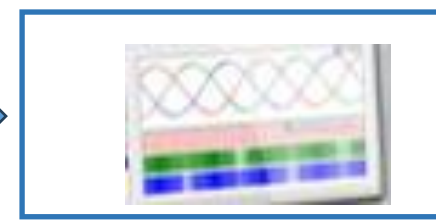
Plant



Control



SCADA





Computation of reference currents

- Active and reactive power in abc coordinates:

$$p = i_a v_a + i_b v_b + i_c v_c \quad q = \frac{1}{\sqrt{3}} [(v_b - v_c)i_a + (v_c - v_a)i_b + (v_a - v_b)i_c]$$

- Matrix formulation in abc coordinates:

$$p = \mathbf{i}_{abc}^T \mathbf{v}_{abc} \quad q = \frac{1}{\sqrt{3}} [\mathbf{i}_{abc}^T \mathbf{M} \mathbf{v}_{abc}] \quad \mathbf{M} = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

- Active and reactive power in dq coordinates:

$$p = (\mathbf{T}_p \mathbf{i}_{dq})^T \mathbf{T}_p \mathbf{v}_{dq} \quad q = \frac{1}{\sqrt{3}} [(\mathbf{T}_p \mathbf{i}_{dq})^T \mathbf{M} \mathbf{T}_p \mathbf{v}_{dq}]$$

$$p = \frac{3}{2} (v_d i_d - v_q i_q) \quad q = \frac{3}{2} (v_d i_q - v_q i_d)$$





Computation of reference currents

- Active and reactive power in dq coordinates (matrix form):

$$p = \frac{3}{2}(-v_d i_d + v_q i_q) \qquad q = \frac{3}{2}(v_d i_q - v_q i_d)$$

$$\begin{bmatrix} p \\ q \end{bmatrix} = \frac{3}{2} \begin{bmatrix} -v_d & v_q \\ -v_q & v_d \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \rightarrow \begin{bmatrix} i_d \\ i_q \end{bmatrix} = \frac{2}{3} \frac{1}{v_d^2 + v_q^2} \begin{bmatrix} v_d & -v_q \\ v_q & -v_d \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix}$$

- Current references in dq coordinates:

$$i_d^* = \frac{2}{3} \left(\frac{p^* v_{sd} - q^* v_{sq}}{v_{sd}^2 + v_{sq}^2} \right) \qquad i_q^* = \frac{2}{3} \left(\frac{p^* v_{sq} - q^* v_{sd}}{v_{sd}^2 + v_{sq}^2} \right)$$

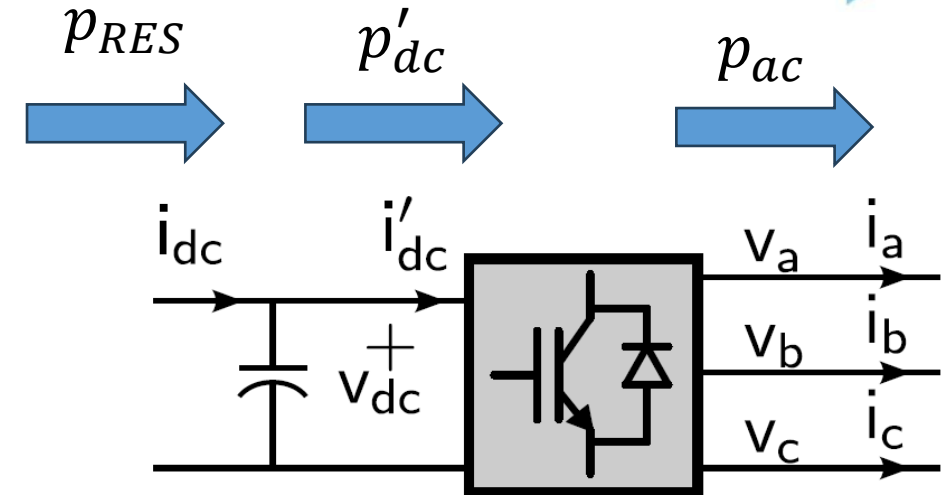


DC-link voltage control

- DC-side equivalent circuit in abc and dq:

$$C \frac{dv_{dc}}{dt} + i_{dc} = i'_{dc} = \frac{1}{2} (\eta_a i_a + \eta_b i_b + \eta_c i_c)$$

$$C \frac{dv_{dc}}{dt} + i_{dc} = i'_{dc} = \frac{1}{2} (\eta_d i_d + \eta_q i_q)$$



- Power balance between AC and DC side of the VSC

$$p'_{dc} \approx p_{ac} \quad \longrightarrow \quad v_{dc} i'_{dc} \approx p_{ac} \quad \longrightarrow \quad v_{dc} \left(C \frac{dv_{dc}}{dt} + i_{dc} \right) \approx p_{ac}$$

$$\frac{C}{2} \frac{dv_{dc}^2}{dt} + v_{dc} i_{dc} \approx p_{ac} \quad \longrightarrow \quad s \frac{C}{2} v_{dc}^2 i'_{dc} + p_{RES} \approx p_{ac}$$



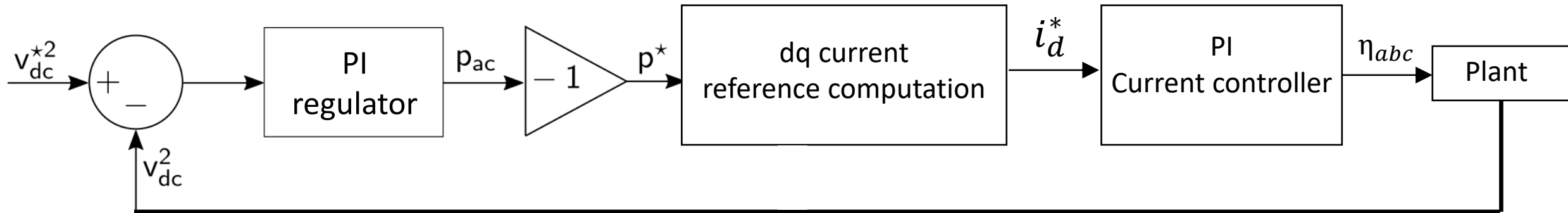


DC-link voltage control

- PI regulator provides the active power reference to maintain v_{dc}^2 :

$$s \frac{C}{2} v_{dc}^2 i'_{dc} + \boxed{p_{RES}} \approx p_{ac}$$

Perturbation $\rightarrow$



Methodology



Study of the state of the art
✓ and development

Modelling and validation in
✓ off-line simulation

Modelling and validation in
Control Hardware in the Loop

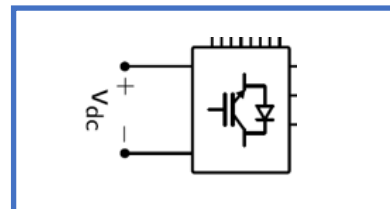
Validation in
Experimental Platform

Real Time Simulation

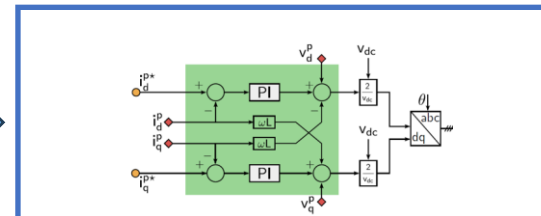
Real-Time Simulator



Host
PC



Plant



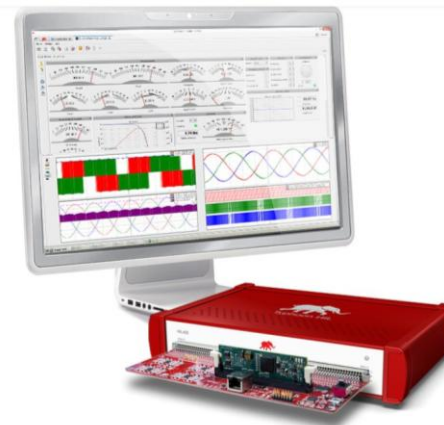
Control



SCADA



Methodology



Study of the state of the art
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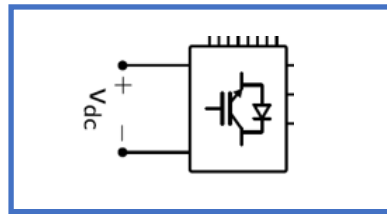
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C-HIL

Real-Time Simulator



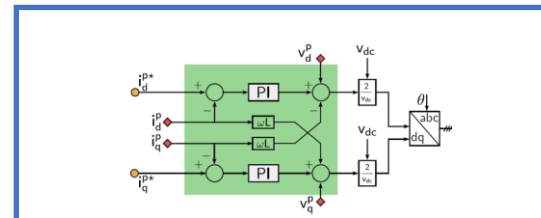
Plant



DSP



Host
PC



Control



SCADA





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